Лабораторная работа №1

Подготовка лабораторного стенда

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Информация

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Цели и задачи

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- Целью данной работы является приобретение практических навыков установки Rocky Linux на виртуальную машину с помощью инструмента Vagrant.

1. Сформируйте box-файл с дистрибутивом Rocky Linux для VirtualBox (см. раздел 1.5.2 или 1.5.3).
2. Запустите виртуальные машины сервера и клиента и убедитесь в их работоспособности.
3. Внесите изменения в настройки загрузки образов виртуальных машин server и client, добавив пользователя с правами администратора и изменив названия хостов (см. раздел 1.5.4).
4. Скопируйте необходимые для работы с Vagrant файлы и box-файлы виртуальных машин на внешний носитель. Используя эти файлы, вы можете попробовать развернуть виртуальные машины на другом компьютере.

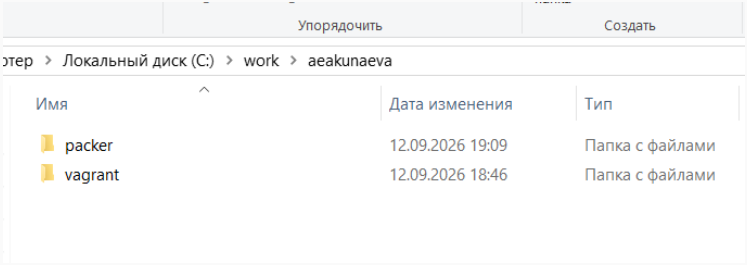
Материалы и методы

Материалы и методы

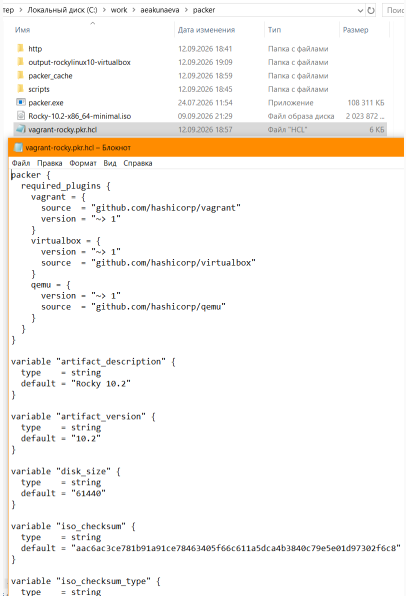
- Linux (дистрибутив Rocky 10.2)
- Linux Fedora Workstation (Markdown)
- Oracle VirtualBox 7.2.16
- Vagrant 2.4.9 (Windows amd64)
- FAR Manager 3.0 build 6699 x64
- Packer 1.16.0 (Windows amd64)

Выполнение лабораторной работы

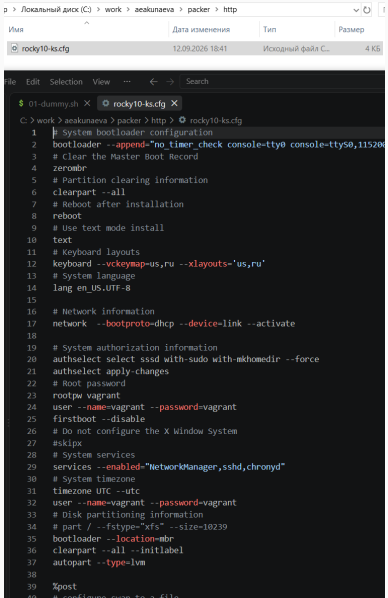
Создание каталога для проекта



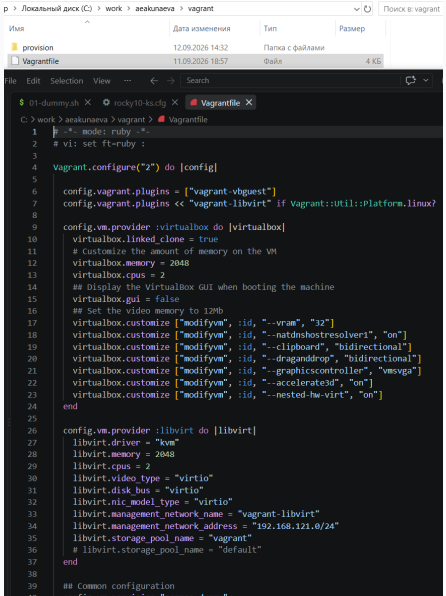
Добавление файла vagrant-rocky.pkr.hcl



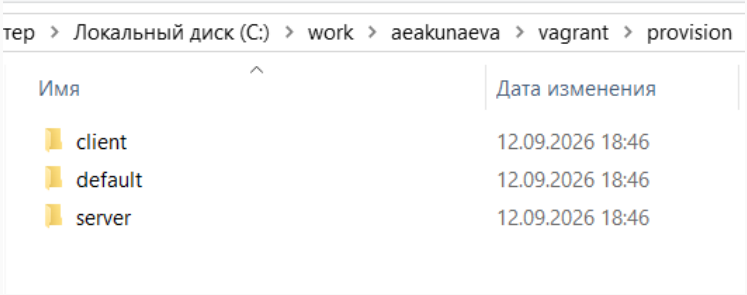
Файл packer/http/rocky10-ks.cfg



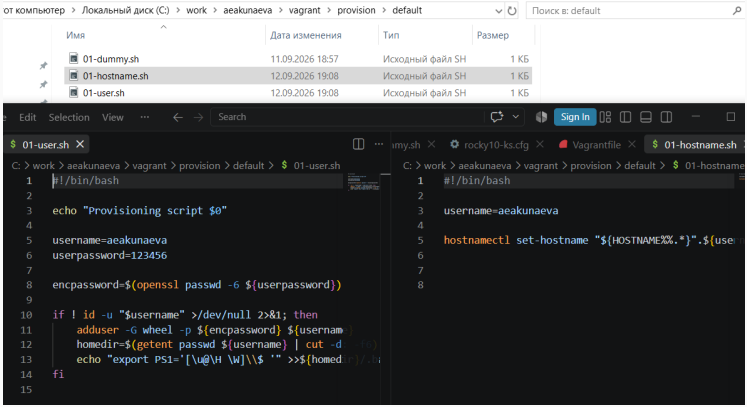
Файл vagrant/Vagrantfile



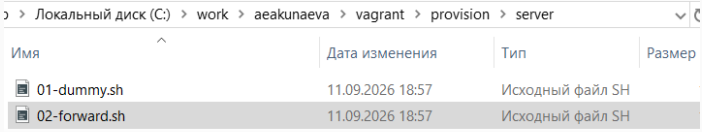
Подкаталоги vagrant/provision/client default server



Файлы подкаталога vagrant/provision/default



Файлы подкаталога vagrant/provision/server



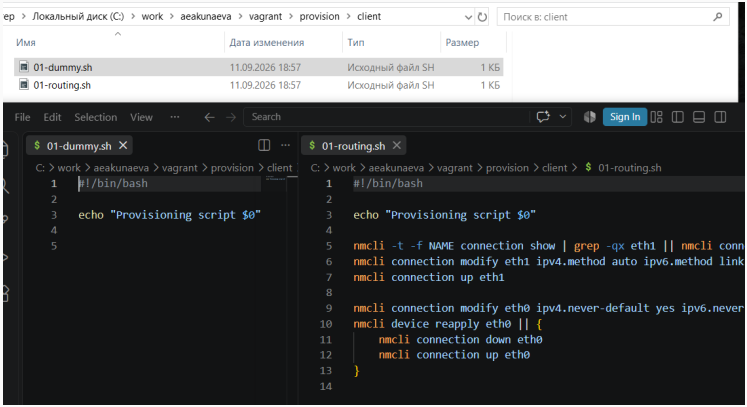
File Edit Selection View ··· Search

02-forward.sh

C: > work > aeakunaeva > vagrant > provision > server > 02-forward.sh

```
1  #!/bin/bash
2
3  echo "Provisioning script $0"
4
5  echo "Enable forwarding"
6  echo "net.ipv4.ip_forward = 1" > /etc/sysctl.d/90-forward.conf
7  sysctl -w net.ipv4.ip_forward=1
8
9  echo "Configure masquerading"
10 firewall-cmd --add-masquerade --permanent
11 firewall-cmd --reload
12
13 restorecon -vR /etc
14
```

Файлы подкаталога vagrant/provision/client



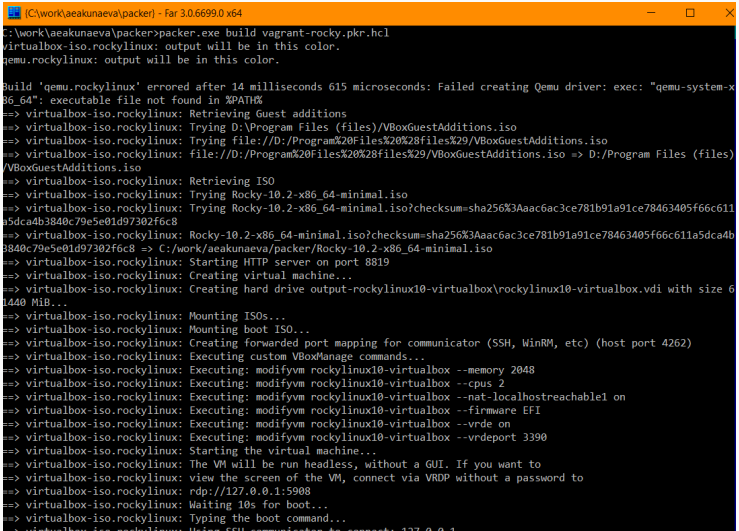
Установка Rocky Linux при помощи FAR: init

```
packer.exe init vagrant-rocky.pkr.hcl
```

```
C:\work\aeakunaeva\packer>packer.exe init vagrant-rocky.pkr.hcl
Installed plugin github.com/hashicorp/vagrant v1.1.7 in "C:/Users/Erdni/AppData/Roaming/packer.d/plugins/github.com/hashi
corp/vagrant/packer-plugin-vagrant_v1.1.7_x5.0_windows_amd64.exe"
Installed plugin github.com/hashicorp/virtualbox v1.1.5 in "C:/Users/Erdni/AppData/Roaming/packer.d/plugins/github.com/ha
shicorp/virtualbox/packer-plugin-virtualbox_v1.1.5_x5.0_windows_amd64.exe"
Installed plugin github.com/hashicorp/qemu v1.1.6 in "C:/Users/Erdni/AppData/Roaming/packer.d/plugins/github.com/hashicor
p/qemu/packer-plugin-qemu_v1.1.6_x5.0_windows_amd64.exe"
```

Установка Rocky Linux при помощи FAR: build

```
packer.exe build vagrant-rocky.pkr.hcl
```



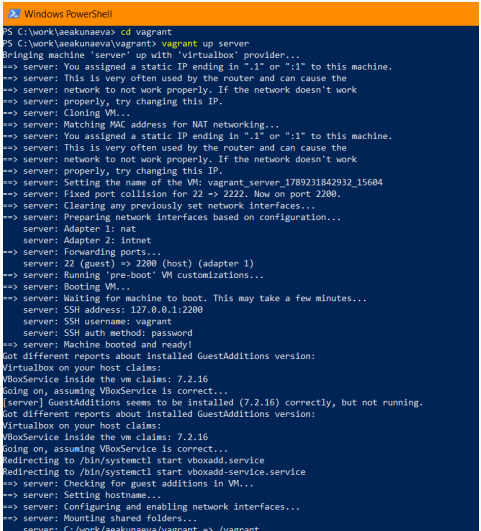
Регистрация образа виртуальной машины

```
vagrant box add rockylinux10 vagrant-virtualbox-rockylinux10-x86_64.box
```

```
C:\work\aeakunaeva\packer>vagrant box add rockylinux10 vagrant-virtualbox-rockylinux10-x86_64.box
==> box: Box file was not detected as metadata. Adding it directly...
==> box: Adding box 'rockylinux10' (v0) for provider: (amd64)
    box: Unpacking necessary files from: file://C:/work/aeakunaeva/packer/vagrant-virtualbox-rockylinux10-x86_64.box
    box:
==> box: Successfully added box 'rockylinux10' (v0) for '(amd64)'!
```

Запуск виртуальной машины Server

```
vagrant up server
```



Запуск виртуальной машины Client

```
vagrant up client
```

```
PS C:\work\aeakunaeva\vagrant> vagrant up client
Bringing machine 'client' up with 'virtualbox' provider...
==> client: Running provisioner: common dummy (shell)...
    client: Running: C:/Users/Erdni/AppData/Local/Temp/vagrant-shell20260912-15672-yc8aoz.sh
    client: Provisioning script /tmp/vagrant-shell
==> client: Running provisioner: common hostname (shell)...
    client: Running: C:/Users/Erdni/AppData/Local/Temp/vagrant-shell20260912-15672-y3yn1i.sh
==> client: Running provisioner: common user (shell)...
    client: Running: C:/Users/Erdni/AppData/Local/Temp/vagrant-shell20260912-15672-isokxq.sh
    client: Provisioning script /tmp/vagrant-shell
==> client: Running provisioner: client dummy (shell)...
    client: Running: C:/Users/Erdni/AppData/Local/Temp/vagrant-shell20260912-15672-g70ueh.sh
    client: Provisioning script /tmp/vagrant-shell
==> client: Running provisioner: client routing (shell)...
    client: Running: C:/Users/Erdni/AppData/Local/Temp/vagrant-shell20260912-15672-b9mws0.sh
    client: Provisioning script /tmp/vagrant-shell
    client: Error: Failed to modify connection 'eth1': failed to update connection: error writing to file '/etc/NetworkM
anager/system-connections/eth1.nmconnection': failed rename /etc/NetworkManager/system-connections/eth1.nmconnection.2S7
IV3 to /etc/NetworkManager/system-connections/eth1.nmconnection: Permission denied
    client: Error: Connection activation failed: IP configuration could not be reserved (no available address, timeout,
etc.)
    client: Hint: use 'journalctl -xe NM_CONNECTION=d95c68da-f55c-41f5-b2b3-cc5c1bf04699 + NM_DEVICE=eth1' to get more d
etails.
    client: Connection successfully reapplied to device 'eth0'.
PS C:\work\aeakunaeva\vagrant>
```

Подключение к серверу и клиенту из консоли и выход

```
vagrant ssh server/client
su - aeakunaeva
exit
```

```
PS C:\work\aeakunaeva\vagrant> vagrant ssh server
==> server: The machine you're attempting to SSH into is configured to use
==> server: password-based authentication. Vagrant can't script entering the
==> server: password for you. If you're prompted for a password, please enter
==> server: the same password you have configured in the Vagrantfile.
vagrant@127.0.0.1's password:
Last login: Sat Sep 12 16:27:29 2026 from 10.0.2.2
vagrant@server: $ su - aeakunaeva
Password:
[aeakunaeva@server.aeakunaeva.net ~]$ exit
logout
vagrant@server: $ vagrant ssh client
bash: vagrant: command not found...
exit
^C
vagrant@server: $ exit
logout
PS C:\work\aeakunaeva\vagrant> vagrant ssh client
==> client: The machine you're attempting to SSH into is configured to use
==> client: password-based authentication. Vagrant can't script entering the
==> client: password for you. If you're prompted for a password, please enter
==> client: the same password you have configured in the Vagrantfile.
vagrant@127.0.0.1's password:
Last login: Sat Sep 12 16:27:29 2026 from 10.0.2.2
vagrant@client: $ su - aeakunaeva
```

Проверка файла Vagrantfile

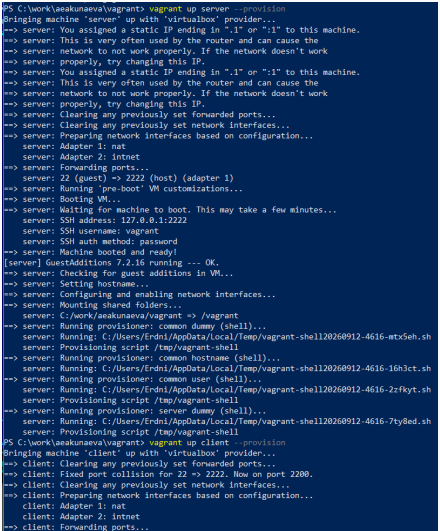
Vagrantfile

C: > work > aeakunaeva > vagrant > Vagrantfile

```
4   Vagrant.configure("2") do |config|
39    ## Common configuration
40    config.vm.provision "common dummy",
41                        type: "shell",
42                        preserve_order: true,
43                        path: "provision/default/01-dummy.sh"
44
45    config.vm.provision "common hostname",
46                        type: "shell",
47                        preserve_order: true,
48                        run: "always",
49                        path: "provision/default/01-hostname.sh"
50
51    config.vm.provision "common user",
52                        type: "shell",
53                        preserve_order: true,
54                        path: "provision/default/01-user.sh"
55
56    ## Server configuration
57    config.vm.define "server", autostart: false do |server|
```

Сохранение изменений внутренних настроек ВМ для Server

```
vagrant up server --provision
```



Проверка приглашения для Server

```
PS C:\work\aeakunaeva\vagrant> vagrant ssh server
==> server: The machine you're attempting to SSH into is configured to use
==> server: password-based authentication. Vagrant can't script entering the
==> server: password for you. If you're prompted for a password, please enter
==> server: the same password you have configured in the Vagrantfile.
vagrant@127.0.0.1's password:
Last login: Sat Sep 12 16:58:27 2026 from 10.0.2.2
vagrant@server:~$ su - aeakunaeva
Password:
Last login: Sat Sep 12 16:58:41 UTC 2026 on pts/0
[aeakunaeva@server.aeakunaeva.net ~]$ exit
logout
vagrant@server:~$ exit
logout
```

Сохранение изменений внутренних настроек ВМ для Client

`vagrant up client --provision`

```
PS C:\work\aeakunaeva\vagrant> vagrant up client --provision
Bringing machine 'client' up with 'virtualbox' provider...
==> client: Running provisioner: common dummy (shell)...
    client: Running: C:/Users/Erdni/AppData/Local/Temp/vagrant-shell120260912-14164-mgbinq.sh
    client: Provisioning script /tmp/vagrant-shell
==> client: Running provisioner: common hostname (shell)...
    client: Running: C:/Users/Erdni/AppData/Local/Temp/vagrant-shell120260912-14164-wxgy2m.sh
==> client: Running provisioner: common user (shell)...
    client: Running: C:/Users/Erdni/AppData/Local/Temp/vagrant-shell120260912-14164-8aggvw.sh
    client: Provisioning script /tmp/vagrant-shell
==> client: Running provisioner: client dummy (shell)...
    client: Running: C:/Users/Erdni/AppData/Local/Temp/vagrant-shell120260912-14164-d1p0ss.sh
    client: Provisioning script /tmp/vagrant-shell
==> client: Running provisioner: client routing (shell)...
    client: Running: C:/Users/Erdni/AppData/Local/Temp/vagrant-shell120260912-14164-q10nrv.sh
    client: Provisioning script /tmp/vagrant-shell
    client: Error: Failed to modify connection 'eth1': failed to update connection: error writing to file '/etc/NetworkM
anager/system-connections/eth1.nmconnection': failed rename /etc/NetworkManager/system-connections/eth1.nmconnection.C7G
PV3 to /etc/NetworkManager/system-connections/eth1.nmconnection: Permission denied
    client: Error: Connection activation failed: IP configuration could not be reserved (no available address, timeout,
etc.)
    client: Hint: use 'journalctl -xe NM_CONNECTION=1bc956e1-7561-401e-8b5d-9bb98a8ffdc9 + NM_DEVICE=eth1' to get more d
etails.
    client: Connection successfully reapplied to device 'eth0'.
PS C:\work\aeakunaeva\vagrant> vagrant ssh client
==> client: The machine you're attempting to SSH into is configured to use
==> client: password-based authentication. Vagrant can't script entering the
==> client: password for you. If you're prompted for a password, please enter
==> client: the same password you have configured in the Vagrantfile.
vagrant@127.0.0.1's password:
Last login: Sat Sep 12 17:08:08 2026 from 10.0.2.2
vagrant@client: $ su - aeakunaeva
Password:
Last login: Sat Sep 12 17:08:14 UTC 2026 on pts/0
[aeakunaeva@client.aeakunaeva.net ~]$ exit
logout
vagrant@client: $ exit
logout
```

Выводы

Выводы

Я приобрела практические навыки установки Rocky Linux на виртуальную машину с помощью инструмента Vagrant.